# Week 9

**Course Level Objectives:**

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Perform hypothesis testing using statistical methods. (CLO 8)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** Describe how a randomization distribution is generated assuming $H_0$ is true. (CLO 5, CLO 8)<br>**2.** Identify the center of a randomization distribution as the null value. (CLO 5)<br>**3.** Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5)<br>**4.** Compute a p-value from a randomization distribution. (CLO 3, CLO 8)<br>**5.** Determine the correct tail(s) for a p-value from the alternative hypothesis. (CLO 8)<br>**6.** Interpret a p-value in context. (CLO 4, CLO 8) | e-Text: 4.2 Measuring Evidence with P-Values<br><br>Interactive Lecture: 4.2 Measuring Evidence with P-Values<br><br>Slides: 4.2 Measuring Evidence with P-Values<br><br>StatKey – Randomization Hypothesis Tests | Participation: Randomization Distributions – StatKey: dolphin study (16 flips); count how many simulated statistics are as extreme as observed<br><br>Your Turn: Tea and Immunity in StatKey (re-randomize tea vs. coffee, find the p-value) | Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM)<br><br>Take-Home Quiz 5 (due Mon Oct 26, 11:59 PM) – covers 4.1–4.3 (all objectives)<br><br>Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives)<br><br>Test 2 Study Guide (all objectives)<br><br>Practice Test 2 (optional; due Thu Oct 29) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Respondus |
| **7.** Relate the size of a p-value to the strength of evidence against $H_0$. (CLO 8)<br>**8.** Compare a p-value to the significance level α to decide whether to reject $H_0$. (CLO 8)<br>**9.** Write a two-step conclusion in context that avoids saying “accept $H_0$”. (CLO 8) | e-Text: 4.3 Determining Statistical Significance<br><br>Interactive Lecture: 4.3 Determining Statistical Significance<br><br>Slides: 4.3 Determining Statistical Significance (includes “What is a p-value?” video) | Participation: P-values and Conclusions – resveratrol examples (p = 0.04, 0.12, 0.0003): strength of evidence, decision at α, conclusion in context<br><br>Your Turn: Decide and Conclude | Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM)<br><br>Take-Home Quiz 5 (due Mon Oct 26, 11:59 PM) – covers 4.1–4.3 (all objectives)<br><br>Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives)<br><br>Test 2 Study Guide (all objectives)<br><br>Practice Test 2 (optional; due Thu Oct 29) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Respondus |